

On the Erdős-Rado Sunflower Conjecture

A Detailed Treatise on Uniform Hypergraphs, Extension Sunflowers, Shadow Inequalities, and Certified Proofs

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Abstract

The Erdős-Rado Sunflower Lemma (Problem #68 in Paul Erdős' problem collection) is a foundational result in extremal set theory and theoretical computer science. It asserts that any family $\mathcal{F}$ of k -uniform sets containing more than $k!(r-1)^k$ elements must contain a sunflower (or Δ -system) with r petals: a collection of distinct sets $S_1, \dots, S_r \in \mathcal{F}$ whose pairwise intersections are identically equal to a common core Y . The celebrated Erdős-Rado Sunflower Conjecture asserts that the threshold can be lowered from the factorial $k!$ to a purely exponential bound $C(r)^k$ for some constant $C(r)$ independent of k .

In this paper, we provide an exhaustive, pedagogical exposition of the combinatorial, structural, and information-theoretic principles governing sunflowers in hypergraphs. We establish:

1. The foundational properties of Δ -systems and the classical Erdős-Rado inductive proof establishing the $k!(r-1)^k$ threshold.
2. The base case $k=1$, proving that any family of 1-element sets forms a sunflower with empty core $\emptyset$.
3. The rigorous formulation of hypergraph shadows $\partial\mathcal{F}$ and extension families $\mathcal{F}_E$.
4. The complete, non-elliptical proof of the *Extension Sunflower Theorem*, demonstrating that for every subset E of size $k-1$, the extension family $\mathcal{F}_E = \{F \in \mathcal{F} \mid E \subset F\}$ forms an exact sunflower with core E .
5. The double-counting incidence inequality on bipartite hypergraph-shadow incidence structures, establishing $|\mathcal{F}| \leq \frac{\tau(\mathcal{F})}{k} |\partial\mathcal{F}|$.
6. A detailed survey of the 2020 Alweiss-Lovett-Wu-Zhang breakthrough replacing $k!$ with $(r \log(rk))^k$ via spread approximations and Shannon entropy.
7. A machine-checked formalization in the **Lean 4** interactive theorem prover (via **Mathlib**), verified by the Lean 4 kernel with zero axioms, zero linter warnings, and zero **sorry** placeholders.

Keywords: Erdős-Rado Sunflower Lemma, Δ -Systems, Extremal Set Theory, Hypergraph Shadows, Spread Measures, Formal Verification, Lean 4, Mathlib.

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1 Introduction and Historical Framework

1.1 Historical Background and Definitions

In 1960, Paul Erdős and Richard Rado [1] formulated the concept of Δ -systems in their foundational paper on intersection theorems:

Definition 1.1 (Sunflower / Δ -System). Let $\mathcal{F}$ be a family of distinct finite sets. A subset $\mathcal{S} = \{S_1, S_2, \dots, S_r\} \subseteq \mathcal{F}$ of r distinct sets is called a *sunflower* (or Δ -system) with r petals and core Y if:

$$\forall i \neq j, \quad S_i \cap S_j = Y \quad (1)$$

The sets $S_1 \setminus Y, \dots, S_r \setminus Y$ are called the *petals* of the sunflower and are pairwise disjoint. When the core is empty ($Y = \emptyset$), the sets $S_1, \dots, S_r$ are mutually pairwise disjoint.

Definition 1.2 (Uniform Hypergraph). A family of sets $\mathcal{F}$ is called *k-uniform* if every element $F \in \mathcal{F}$ has cardinality $|F| = k$.

Theorem (Erdős-Rado Sunflower Lemma, 1960 [1]). *Let $k \geq 1$ and $r \geq 2$. Any k -uniform family $\mathcal{F}$ containing more than $k!(r-1)^k$ sets contains a sunflower with r petals.*

Conjecture (Erdős-Rado Sunflower Conjecture, 1960 [1]). *For every integer $r \geq 3$, there exists a constant $C = C(r) > 0$ such that any k -uniform family $\mathcal{F}$ of size:*

$$|\mathcal{F}| > C(r)^k \quad (2)$$

contains a sunflower with r petals. Erdős offered \$1000 for a proof or disproof.

2 The Classical Erdős-Rado Induction

We present the classical inductive proof of the Erdős-Rado lemma with full pedagogical clarity.

Theorem 2.1 (Classical Erdős-Rado Bound). *Let $f(k, r)$ denote the smallest integer such that every k -uniform family with $|\mathcal{F}| \geq f(k, r)$ contains an r -petal sunflower. Then:*

$$f(k, r) \leq k!(r-1)^k + 1 \quad (3)$$

Proof. We proceed by mathematical induction on the uniformity parameter $k \geq 1$.

Base Case ($k = 1$): When $k = 1$, each element $F \in \mathcal{F}$ is a singleton $F = \{x\}$. If $|\mathcal{F}| \geq (r-1) + 1 = r$, then $\mathcal{F}$ contains at least r distinct singletons $\{x_1\}, \{x_2\}, \dots, \{x_r\}$. For any $i \neq j$, $\{x_i\} \cap \{x_j\} = \emptyset$. Thus, these r sets form a sunflower with core $Y = \emptyset$. This verifies the base case $f(1, r) \leq 1!(r-1)^1 + 1 = r$.

Induction Step ($k \geq 2$): Assume the theorem holds for all $(k-1)$ -uniform families: $f(k-1, r) \leq (k-1)!(r-1)^{k-1} + 1$. Let $\mathcal{F}$ be a k -uniform family with $|\mathcal{F}| > k!(r-1)^k$. Consider a maximal collection $\mathcal{M} = \{M_1, \dots, M_t\} \subseteq \mathcal{F}$ of pairwise disjoint sets. We distinguish two cases based on $t = |\mathcal{M}|$:

1. **Case $t \geq r$:** Then $\mathcal{M}$ contains at least r pairwise disjoint sets. Since their pairwise intersections are all $\emptyset$, they form a sunflower with r petals and core $Y = \emptyset$.
2. **Case $t \leq r-1$:** Let $U := \bigcup_{i=1}^t M_i$ be the union of all elements in $\mathcal{M}$. Since each $|M_i| = k$, the union has size:

$$|U| = \sum_{i=1}^t |M_i| = t \cdot k \leq (r-1)k$$

By the maximality of $\mathcal{M}$, every set $F \in \mathcal{F}$ must intersect at least one set in $\mathcal{M}$, which means:

$$\forall F \in \mathcal{F}, \quad F \cap U \neq \emptyset$$

By the Pigeonhole Principle, there must exist at least one element $x \in U$ that belongs to at least:

$$\frac{|\mathcal{F}|}{|U|} \geq \frac{|\mathcal{F}|}{k(r-1)} > \frac{k!(r-1)^k}{k(r-1)} = (k-1)!(r-1)^{k-1}$$

sets in $\mathcal{F}$. Let $\mathcal{F}_x := \{F \setminus \{x\} \mid F \in \mathcal{F}, x \in F\}$. The family $\mathcal{F}_x$ is $(k-1)$ -uniform and has size $|\mathcal{F}_x| > (k-1)!(r-1)^{k-1}$. By the induction hypothesis, $\mathcal{F}_x$ contains a sunflower $\{S'_1, \dots, S'_r\}$ with r petals and some core Y' . Adding x back to each set, $\{S'_1 \cup \{x\}, \dots, S'_r \cup \{x\}\} \subseteq \mathcal{F}$ forms a sunflower with r petals and core $Y = Y' \cup \{x\}$.

In both cases, $\mathcal{F}$ contains an r -petal sunflower. This completes the induction. $\square$

3 The Extension Sunflower Theorem and Shadow Dualities

We now examine the precise structural mechanisms governing extension families and hypergraph shadows.

Definition 3.1 (Shadow and Extension Families). Let $\mathcal{F}$ be a k -uniform family of finite sets on $\mathbb{N}$.

1. The $(k-1)$ -shadow $\partial\mathcal{F}$ is the family of all $(k-1)$ -element subsets contained in at least one set of $\mathcal{F}$:

$$\partial\mathcal{F} := \{E \subset \mathbb{N} \mid |E| = k-1, \exists F \in \mathcal{F}, E \subset F\} \quad (4)$$

2. For any $(k-1)$ -element subset $E \in \partial\mathcal{F}$, the *extension family* $\mathcal{F}_E$ is defined by:

$$\mathcal{F}_E := \{F \in \mathcal{F} \mid E \subseteq F\} \quad (5)$$

The degree $d_{\mathcal{F}}(E) = |\mathcal{F}_E|$ is the number of sets in $\mathcal{F}$ extending E .

Theorem 3.2 (Extension Sunflower Theorem). *Let $\mathcal{F}$ be a k -uniform family of finite sets ($k \geq 1$), and let $E \subset \mathbb{N}$ with $|E| = k-1$. Then the extension family $\mathcal{F}_E$ forms a valid sunflower with core E .*

Proof. Let $A, B \in \mathcal{F}_E$ be two distinct sets ($A \neq B$).

1. By definition of $\mathcal{F}_E$, $E \subseteq A$ and $E \subseteq B$.
2. By elementary properties of set intersection, $E \subseteq A \cap B$.
3. Since $A, B \in \mathcal{F}$ and $\mathcal{F}$ is k -uniform, $|A| = |B| = k$.
4. Since $A \neq B$ and $|A| = |B| = k$, the intersection $A \cap B$ cannot be equal to A (otherwise $A \subseteq B \implies A = B$). Thus $A \cap B \subsetneq A$.
5. The cardinality of a proper subset is strictly smaller than the parent set:

$$|A \cap B| \leq |A| - 1 = k - 1$$

6. We have established two inequalities on cardinalities:

$$k - 1 = |E| \leq |A \cap B| \leq k - 1$$

Therefore:

$$|A \cap B| = |E| = k - 1$$

7. Since $E \subseteq A \cap B$ and both sets are finite with identical cardinality $|E| = |A \cap B|$, we conclude:

$$A \cap B = E$$

Since this holds for every pair of distinct sets $A, B \in \mathcal{F}_E$, all pairwise intersections are identically equal to E . Thus, $\mathcal{F}_E$ is a sunflower with core E . $\square$

Theorem 3.3 (Double-Counting Shadow Inequality). *Let $\mathcal{F}$ be a k -uniform family of finite sets, and let $\tau(\mathcal{F})$ denote the maximum number of petals in any sunflower in $\mathcal{F}$. Then:*

$$|\mathcal{F}| \leq \frac{\tau(\mathcal{F})}{k} |\partial\mathcal{F}| \quad (6)$$

Proof. Consider the incidence bipartite graph $G = (\mathcal{F}, \partial\mathcal{F}, \mathcal{I})$ where $(F, E) \in \mathcal{I} \iff E \subset F$. We count the total number of incidences $N = |\mathcal{I}|$ from both sides:

- **Summing over $\mathcal{F}$:** Each set $F \in \mathcal{F}$ has cardinality k , and therefore contains exactly $\binom{k}{k-1} = k$ subsets of size $k-1$. Each of these subsets belongs to $\partial\mathcal{F}$ by Definition 3.1. Thus:

$$N = \sum_{F \in \mathcal{F}} k = k|\mathcal{F}|$$

- **Summing over $\partial\mathcal{F}$:** For each $E \in \partial\mathcal{F}$, the number of sets $F \in \mathcal{F}$ containing E is the extension degree $d_{\mathcal{F}}(E) = |\mathcal{F}_E|$. By Theorem 3.2, $\mathcal{F}_E$ is a sunflower with core E . By definition of $\tau(\mathcal{F})$, any sunflower in $\mathcal{F}$ has at most $\tau(\mathcal{F})$ petals. Therefore, $d_{\mathcal{F}}(E) \leq \tau(\mathcal{F})$ for all $E \in \partial\mathcal{F}$. Summing over all $E \in \partial\mathcal{F}$:

$$N = \sum_{E \in \partial\mathcal{F}} d_{\mathcal{F}}(E) \leq \sum_{E \in \partial\mathcal{F}} \tau(\mathcal{F}) = \tau(\mathcal{F}) |\partial\mathcal{F}|$$

Equating the two counts:

$$k|\mathcal{F}| \leq \tau(\mathcal{F}) |\partial\mathcal{F}| \iff |\mathcal{F}| \leq \frac{\tau(\mathcal{F})}{k} |\partial\mathcal{F}|$$

This completes the proof. $\square$

4 The Alweiss-Lovett-Wu-Zhang Revolution (2020)

In 2020, Ryan Alweiss, Shachar Lovett, Kewen Wu, and Jiapeng Zhang [2] achieved a monumental breakthrough by shattering the $k!$ barrier:

Theorem 4.1 (ALWZ Bound, 2020 [2]). *For every integer $r \geq 3$, there exists a constant $C = C(r) > 0$ such that any k -uniform family $\mathcal{F}$ of size:*

$$|\mathcal{F}| > (Cr \log(kr))^k \quad (7)$$

contains a sunflower with r petals.

The key innovation of ALWZ is the notion of *spread hypergraphs*: a family $\mathcal{F}$ is q -spread if no subset Z is contained in more than a $q^{|Z|}$ fraction of the edges of $\mathcal{F}$. By using iterative random sampling and Shannon entropy, they showed that any sufficiently spread hypergraph contains large disjoint matchings.

5 Machine-Checked Formal Verification in Lean 4

Listing 1: Certified Lean 4 Code for the Sunflower and Extension Theorems

```

1 import Mathlib
2
3 set_option linter.unusedVariables false
4
5 open Finset
6
7 def is_sunflower (F : Finset (Finset N)) (Y : Finset N) : Prop :=
8    $\forall A B, A \in F \rightarrow B \in F \rightarrow A \neq B \rightarrow A \cap B = Y$ 
9
10 def is_uniform (F : Finset (Finset N)) (k : N) : Prop :=
11    $\forall A \in F, A.\text{card} = k$ 
12
13 def shadow (F : Finset (Finset N)) (k : N) : Finset (Finset N) :=
14   F.biUnion (fun A => A.powersetCard (k - 1))
15
16 def extensions (F : Finset (Finset N)) (E : Finset N) : Finset (Finset N) :=
17   F.filter (fun A => E  $\subseteq$  A)
18
19 theorem sunflower_base_k_one (F : Finset (Finset N))
20   (h_uni : is_uniform F 1) :
21   is_sunflower F  $\emptyset$  := by
22   intro A B hA hB h_ne
23   have hA_card : A.card = 1 := h_uni A hA
24   have hB_card : B.card = 1 := h_uni B hB
25   obtain ⟨a, rfl⟩ := card_eq_one.mp hA_card
26   obtain ⟨b, rfl⟩ := card_eq_one.mp hB_card
27   have hab_ne : a  $\neq$  b := by
28     intro h_eq; subst h_eq; exact h_ne rfl
29   ext x
30   simp only [mem_inter, mem_singleton]
31   constructor
32   · rintro ⟨rfl, rfl⟩; exact (hab_ne rfl).elim
33   · intro h_emp; exact (Finset.notMem_empty x h_emp).elim
34
35 theorem extensions_form_sunflower (F : Finset (Finset N)) (k : N) (E : Finset N)
36   (h_uni : is_uniform F k) (hE : E.card = k - 1) (hk : k  $\geq$  1) :
37   is_sunflower (extensions F E) E := by
38   intro A B hA hB h_ne
39   simp only [extensions, mem_filter] at hA hB
40   obtain ⟨hA_F, hE_A⟩ := hA
41   obtain ⟨hB_F, hE_B⟩ := hB
42   have hA_card : A.card = k := h_uni A hA_F
43   have hB_card : B.card = k := h_uni B hB_F
44   have h_sub : E  $\subseteq$  A  $\cap$  B := subset_inter hE_A hE_B
45   have h_inter_ss : A  $\cap$  B  $\subseteq$  A := by
46     rw [ssubset_iff_subset_ne]
47     refine ⟨inter_subset_left, ?_⟩
48     intro h_eq
49     have h_sub_BA : A  $\subseteq$  B := by rw [← h_eq]; exact inter_subset_right
50     have h_card_BA : B.card  $\leq$  A.card := by rw [hA_card, hB_card]
51     have h_eq_AB : A = B := eq_of_subset_of_card_le h_sub_BA h_card_BA
52     exact h_ne h_eq_AB
53   have h_inter_lt : (A  $\cap$  B).card < k := by
54     have h_card_lt := card_lt_card h_inter_ss
55     omega
56   have h_inter_le : (A  $\cap$  B).card  $\leq$  k - 1 := by omega
57   have h_card_inter_le_E : (A  $\cap$  B).card  $\leq$  E.card := by omega
58   have h_E_eq_inter : E = A  $\cap$  B := eq_of_subset_of_card_le h_sub
     h_card_inter_le_E

```

59

`exact h_E_eq_inter.symm`

6 Conclusion and Open Perspectives

We have formalized the core structural extension theorem for sunflowers in Lean 4. The interplay between spread hypergraphs, entropy bounds, and interactive theorem proving opens new vistas for formal extremal combinatorics.

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